Question			Answer	Marks	AOs	Guidance	
11	(a)		Line joining (2, 3) to (10, 200) has gradient $\frac{200-3}{10-2}=24.625$	M1	3.3	Attempt to find the gradient of the line	
			Equation of the line is $N=24.625t-46.25$				
			$m=\frac{197}{8}=24.625,\ c=-\frac{185}{4}=-46.25$	A1 A1	1.1 1.1	Each value correct. Accept rounded to 2 or more s.f.	Values may be implied by correct equation
			Alternative solution 1				
			Regression line (BC) for data set (2, 3), (10, 200)	M1		Using the calculator to find the regression line (may be implied)	
			Regression line is $N=24.625t-46.25$				
			$m=24.625,\ c=-46.25$	A1 A1		Each value correct. Accept rounded to 2 or more s.f.	Values may be implied by correct equation
			Alternative solution 2				
			Simultaneous equations				
			Using (2, 3) $2m+c=3$	M1		Attempt to find and solve simultaneous equations (BC)	
			and using (10, 200) $10m+c=200$				
			$m=\frac{197}{8}=24.625,\ c=-\frac{185}{4}=-46.25$	A1 A1		Each value correct. Accept rounded to 2 or more s.f.	
				[3]			
11	(b)		The number of calls increases by 24.625 per day	B1 [1]	3.4	FT their *m*. Condone missing units	
11	(c)		When $t=1,\ N=-21.625$ and N cannot be negative	B1 [1]	3.5b	Argument from a correct value for N FT their equation	
11	(d)		When $t=2,\ N=e^{0.53\times2}=2.886\approx3$	M1	1.1a	Substitution of both values into the model, soi	
			When $t=10,\ N=e^{0.53\times10}=200.33\approx200$				
			So this is a good model for these two points	E1 [2]	3.4	Conclusion from correct values	

Question			Answer	Marks	AOs	Guidance	
11	(e)		rate of increase $\frac{\mathrm{d}N}{\mathrm{d}t} = 0.53\mathrm{e}^{0.53t}$	M1	1.1a	Using gradient of $\mathrm{e}^{kx} = k\mathrm{e}^{kx}$	
			When $t = 10$, $\frac{\mathrm{d}N}{\mathrm{d}t} = 0.53\mathrm{e}^{0.53\times10}$	M1	1.1a	Substituting $t = 10$ into their expression	
			106 new calls per day	B1 [3]	3.4	cao Allow for gradient found BC or numerical method	